Why are trains so often late?

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Motivation

Figure 1: Traffic management in a large railway company: a difficult job

Overview

Delay propagation
- Introduction
- Our congestion-based framework

Statistical guarantees
- Minimax lower bound
- Convergence of a sparse estimator

Results & perspectives
- Numerical experiments on simulated data
- Numerical experiments on real railway logs
- Future work

Traffic management
- Welcome to Flatland
- Our solution approach

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Context of the study

- Goal: a delay prediction model for the Paris suburban network
- Central challenge with many applications:
  - tactical scheduling
  - real-time regulation
  - passenger information
- See our preprint Dalle and De Castro [4] for more details

What is delay propagation?

- We cannot consider trains individually because they interact through shared resources (infrastructure, driver, rolling stock, etc.)
- Primary delay + resource conflicts $\Longrightarrow$ propagated delays

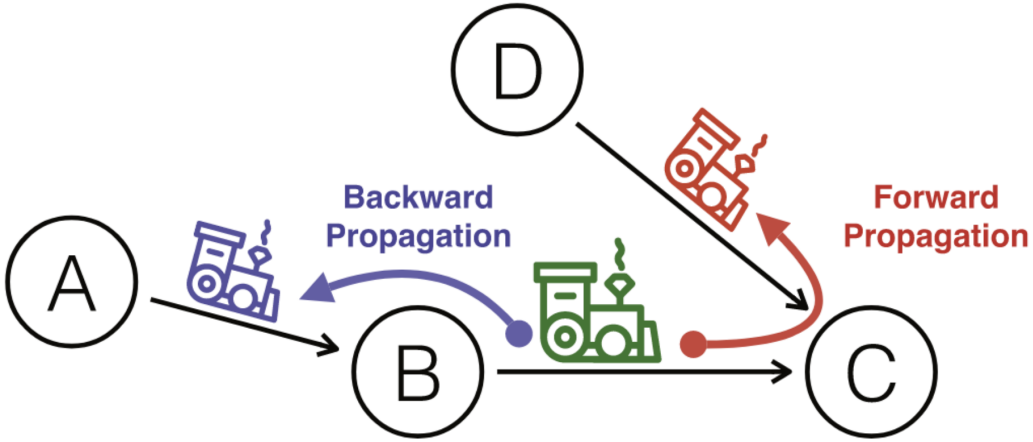

Figure 2: Propagation mechanisms [7]

You are guilty too!

Figure 3: If you're late, please wait for the next train

The time-event graph

- Standard methods assume knowledge of the conflicts to build an event dependency graph and propagate delays

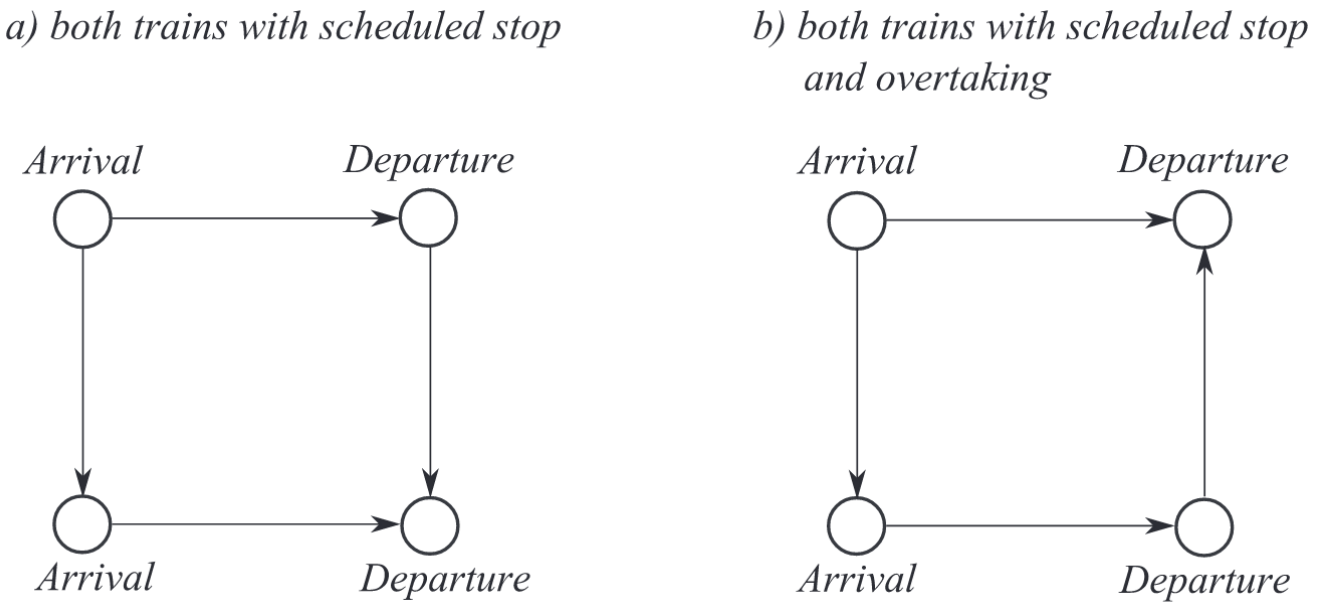

Figure 4: Event dependencies [3]

Why are standard methods insufficient?

- Not enough information available:
  - lack of data on microscopic resource use
  - unpredictable modifications in real time
- Our solution: implicit interactions carried by the railway network

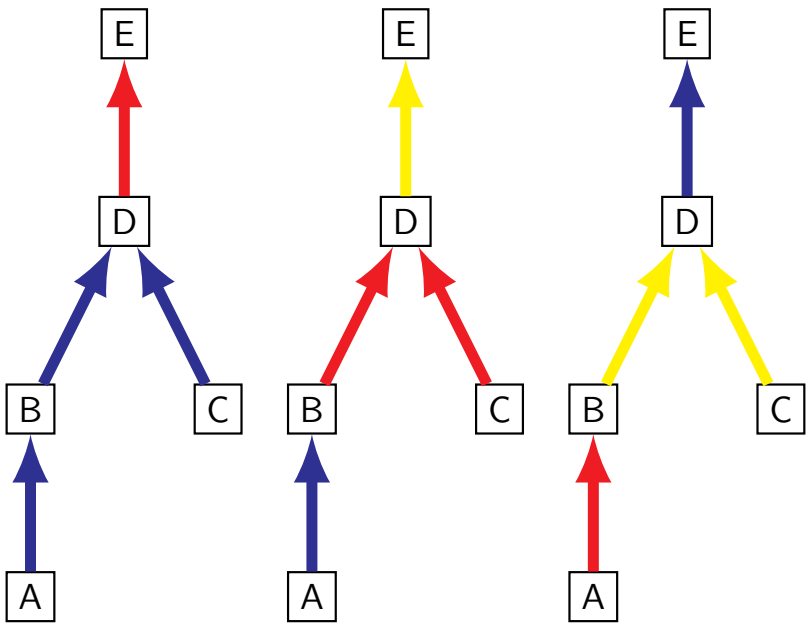

Figure 5: The propagation of congestion

A two-layer propagation model

- A directed graph $G = (V, E)$ for the railway lines and stations
- Latent congestion values $X_{t,e}$ for each time step t:

$$\underbrace{X_{t,e_2}}_{\text{new congestion value}} = \underbrace{\sum_{e_1 \sim e_2} \theta_{e_1,e_2} X_{t-1,e_1}}_{\substack{\text{propagation from} \\ \text{nearby edges}}} + \underbrace{\mathcal{N}(0, \sigma^2)}_{\text{noise}}$$

- Observed arrival times $A_{k,v}$ for each train k depend on the congestion when it crosses the edge (u, v):

$$\underbrace{(A_{k,v} - A^{\text{th}}_{k,v})}_{\text{delay at } v} - \underbrace{(A_{k,u} - A^{\text{th}}_{k,u})}_{\text{delay at } u} = \underbrace{X_{\lfloor A_{k,u}/\Delta t \rfloor,(u,v)}}_{\text{congestion}} + \underbrace{\mathcal{N}(0, \omega^2)}_{\text{noise}}$$

Abstract formulation

- Latent congestion X: Vector AutoRegressive (VAR) process

$$X_t = \theta X_{t-1} + \varepsilon \qquad \text{where} \qquad \varepsilon \sim \mathcal{N}(0, \Sigma)$$

- Observed delay variation $Y = (A_{k,v} - A^{\text{th}}_{k,v}) - (A_{k,u} - A^{\text{th}}_{k,u})$

$$Y = \Pi X + \eta \qquad \text{where} \qquad \eta \sim \mathcal{N}(0, \omega^2 I)$$

Y is a random projection Π of X, where each component (t, e) is present on average p times

Formalizing our assumptions

- Congestion plays a significant role:

$$\mathrm{sp}(\Sigma) \subset [\sigma_{\min}^2, \sigma_{\max}^2] \quad \text{with} \quad \sigma_{\max} \sim \sigma_{\min} \gg \omega$$

- Congestion propagation is local: the transition matrix θ is s-sparse (row-wise) and bounded:

$$\theta \in \Theta_s = \{\theta \in \mathbb{R}^{D \times D} : \|\theta\|_2 \leq \vartheta < 1 \quad \text{and} \quad \forall i, \|\theta_{i,\cdot}\|_0 \leq s\}$$

- Traffic is dense: every edge is crossed often enough ($p \not\ll 1$)
- Temporal & spatial correlations are allowed in the projection Π

Two main questions

1. How hard is it to estimate the transition matrix θ on a specific network?
2. Does this simple model give rise to interesting insights when applied to open railway data?

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Bounding the performance of any estimator

- We study the worst-case probability of error for the best possible estimator:

$$\inf_{\widehat{\theta}} \sup_{\theta \in \Theta_s} \mathbb{P}_\theta(\|\widehat{\theta} - \theta\|_\infty \geq \delta)$$

- Using Fano's method [14], we reduce the problem to finding a set of parameters $(\theta_i)_{0 \leq i \leq M}$ that are both
  1. sufficiently far apart: $\max_{i \neq j} \|\theta_i - \theta_j\|_\infty \geq \alpha$
  2. hard to distinguish in terms of induced distributions:

$$\frac{1}{M+1} \sum_{i=1}^{M} \mathrm{KL}(\mathbb{P}_{\theta_i} \| \mathbb{P}_{\theta_0}) \leq \beta$$

Our negative result

Theorem: Minimax lower bound

Let us define

$$\gamma_\ell = (1-\vartheta)^{3/2}\frac{\sigma_{\min}^2+\omega^2}{\sigma_{\max}^2}.$$

Then there is a universal constant c such that

$$\inf_{\widehat{\theta}}\sup_{\theta\in\Theta_s}\mathbb{P}_\theta\left(\|\widehat{\theta}-\theta\|_\infty \geq c\frac{\gamma_\ell s}{p\sqrt{T}}\right) \geq \frac{1}{2}$$

Key ingredients

Lemma: Covariance decomposition

The covariance matrix of Y given Π decomposes as the sum of a constant term and a residual: $\mathrm{Cov}_\theta[Y|\Pi] = Q_\Pi + R_\Pi(\theta)$.

Lemma: KL divergence between close Gaussians

Let Δ be a symmetric matrix such that $\lambda_{\min}(\Delta) > -1$, and let M be a matrix such that $MM' \succ 0$. Then

$$\mathrm{KL}\left[\mathcal{N}(\mu, M(\mathrm{I}+\Delta)M') \,\middle\|\, \mathcal{N}(\mu, MM')\right] \leq \frac{\|\Delta\|_F^2}{2(1+\lambda_{\min}(\Delta))}.$$

Autocovariance estimation

- The autocovariance of a VAR process satisfies

$$\Gamma_h(\theta) = \mathrm{Cov}_\theta[X_{t+h}, X_t] = \theta^h \Gamma_0(\theta) \qquad \text{(YW)}$$

- We use a scaled and corrected plugin estimator:

$$\widehat{\Gamma}_h := \frac{1}{S(h)} \odot \left[\frac{1}{T-h} \sum_{t=1}^{T-h} \underbrace{\left(\Pi_{t+h}^{+} Y_{t+h}\right)}_{\widehat{X}_{t+h}} \underbrace{\left(\Pi_t^{+} Y_t\right)'}_{\widehat{X}_t'} \right] - C(h).$$

The importance of sparsity

Figure 6: Sparse penalization focuses estimation on a small number of nonzero components

The Dantzig selector

- We look for a sparse approximate solution to (YW). The first option would be the LASSO [13]

$$\widehat{\theta} := \operatorname*{argmin}_{M \in \mathbb{R}^{D^2}} \|M\widehat{\Gamma}_0 - \widehat{\Gamma}_1\|_2^2 + \lambda \|\mathrm{vec}(M)\|_1$$

- We use the Dantzig selector [1] instead (computationally cheaper and theoretically easier)

$$\widehat{\theta} := \operatorname*{argmin}_{M \in \mathbb{R}^{D^2}} \|\mathrm{vec}(M)\|_1 \quad \text{s.t.} \quad \|M\widehat{\Gamma}_0 - \widehat{\Gamma}_1\|_{\max} \leq \lambda \quad \text{(LP)}$$

Our positive result

Theorem: Sparse estimator convergence

Let us define

$$\gamma_u(\theta) = \frac{\|\theta\|_\infty + 1}{(1 - \|\theta\|_2)^2} \frac{\sigma_{\max}^2 + \omega^2}{\|\Gamma_0(\theta)^{-1}\|_1^{-1}}.$$

Then there is a universal constant c such that, for the optimal value of λ, with high probability,

$$\|\widehat{\theta} - \theta\|_\infty \leq c \frac{\gamma_u(\theta) s}{p\sqrt{T}} \sqrt{\log(D/\delta)}$$

Key ingredients

Lemma: Discrete concentration

Let $Z_t = \pi_{t+h,e_1}\pi_{t,e_2}$, where $\pi_{t,e}$ is the number of activations of component (t,e) in the observations. Then

$$\mathbb{P}\Big(\Big|\frac{1}{T-h}\sum_{t=1}^{T-h} Z_t - \mathbb{E}[Z]\Big| \geq u\mathbb{E}[Z]\Big) \leq c_1 \exp(-c_2 u^2 \mathbb{E}[Z]).$$

Requires recent results on Markov chain Chernoff bounds [2] in the non-reversible case

Key ingredients (2)

Lemma: Conditional Gaussian concentration

Let A be a random matrix satisfying $\|A\|_2 \leq M_2$ and $\|A\|_F^2 \leq M_F^2$ with probability $1-\delta$. If $X \sim \mathcal{N}(0, I)$ and $X \perp A$, then

$$\mathbb{P}\left(|X'AX - \mathbb{E}[X'AX]| \geq u\right) \leq \delta + 2\exp\left(-c\min\left\{\frac{u^2}{M_F^2}, \frac{u}{M_2}\right\}\right) + \mathbb{P}\left(|\operatorname{Tr}(A - \mathbb{E}[A])| \geq u/2\right)$$

Based on the Hanson-Wright inequality [15] with a twist (thank you Emeline!)

Comments on the bounds

- Coherent w.r.t. the role of dimension parameters s, p, T
- Influence of the noise $(1+\omega^2/\sigma^2)$ related to Fisher information
- Correlated projections introduce additional challenges:
  - need for custom concentration inequalities
  - mismatch between lower and upper bound in some regards

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Role of the dimension

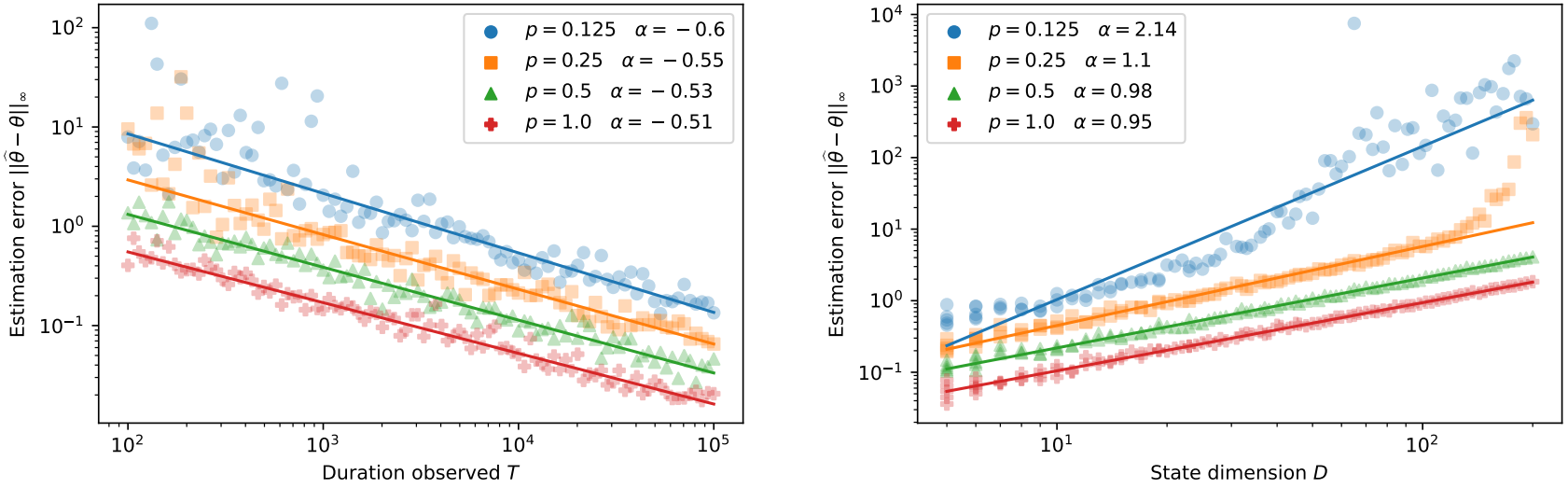

Figure 7: Influence of D and T on the error

Role of the noise

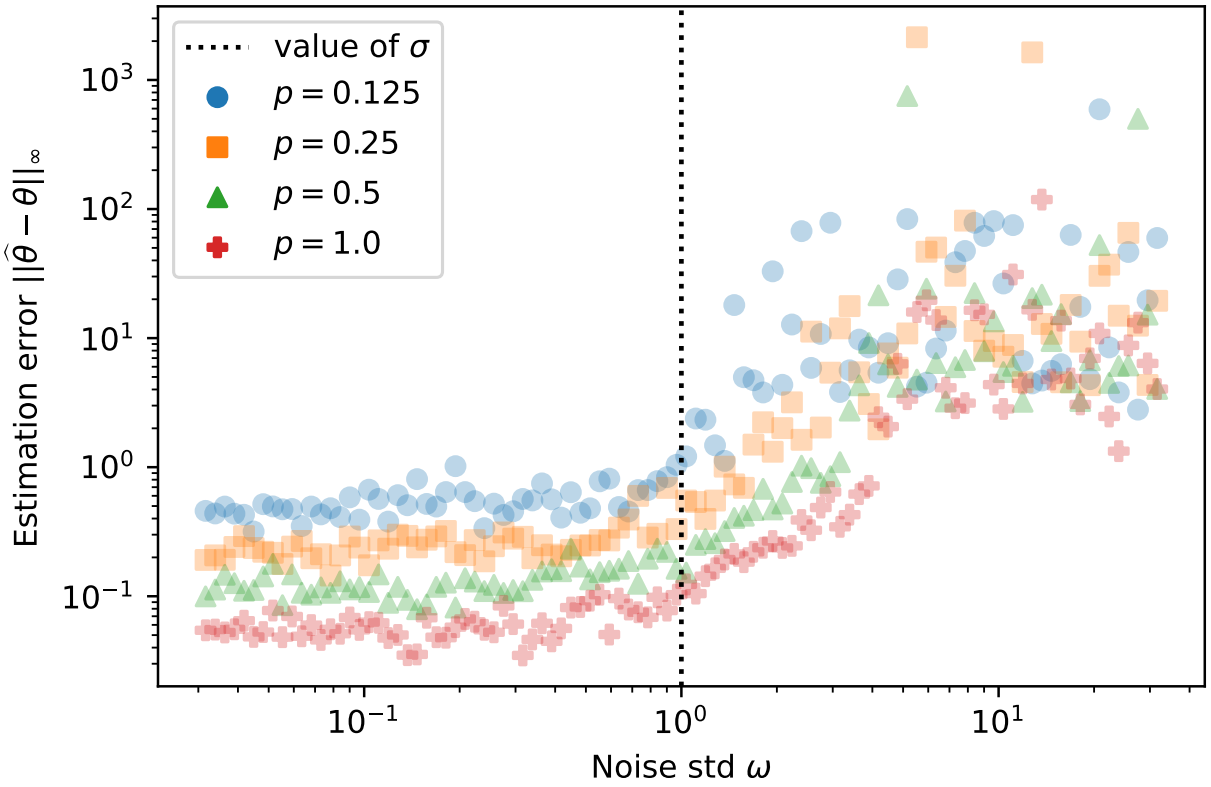

Figure 8: Influence of σ and ω on the error

Role of the sparsity

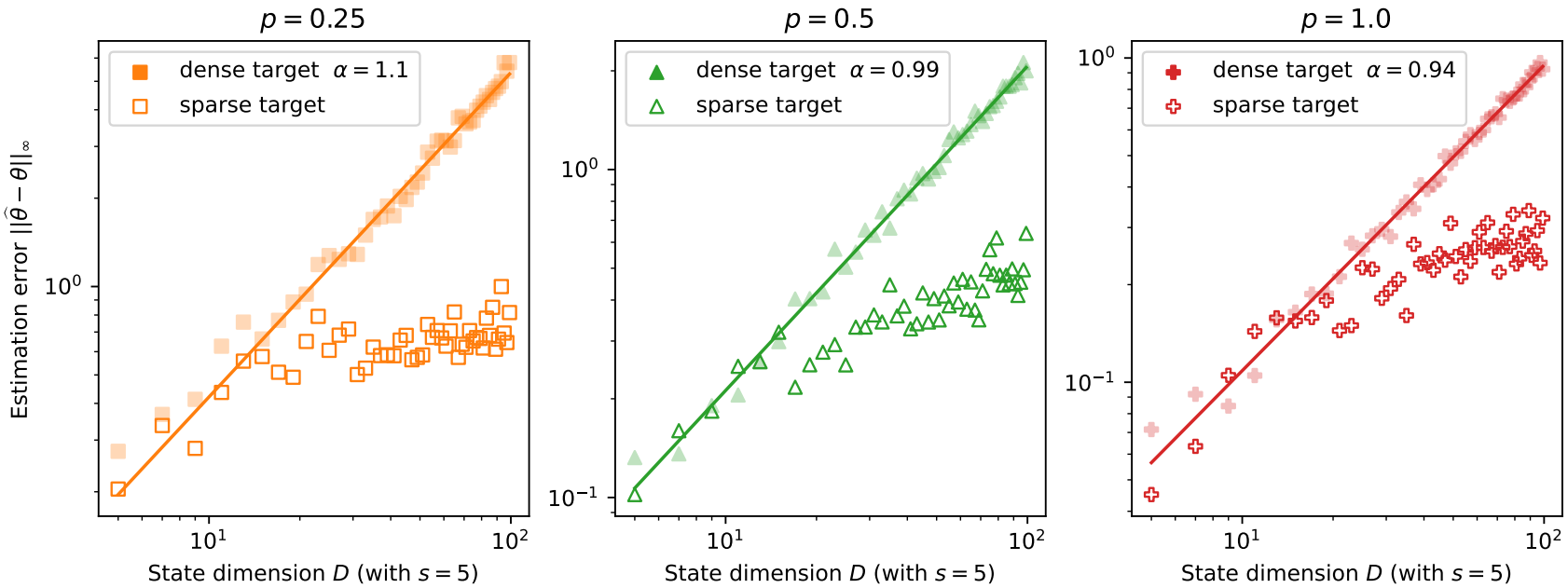

Figure 9: Influence of D on the error with fixed s

Data preprocessing

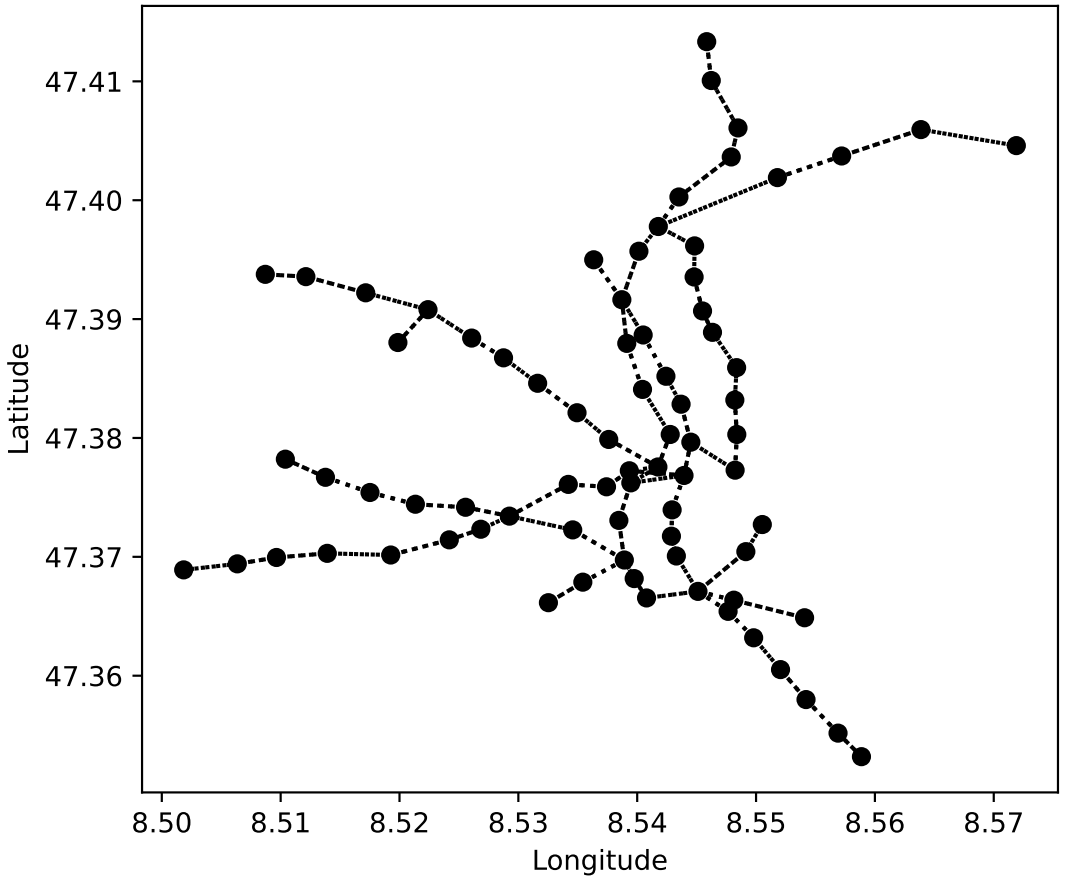

Figure 10: Map of the Zürich tram network generated from consecutive events

Results on Zürich tramway data

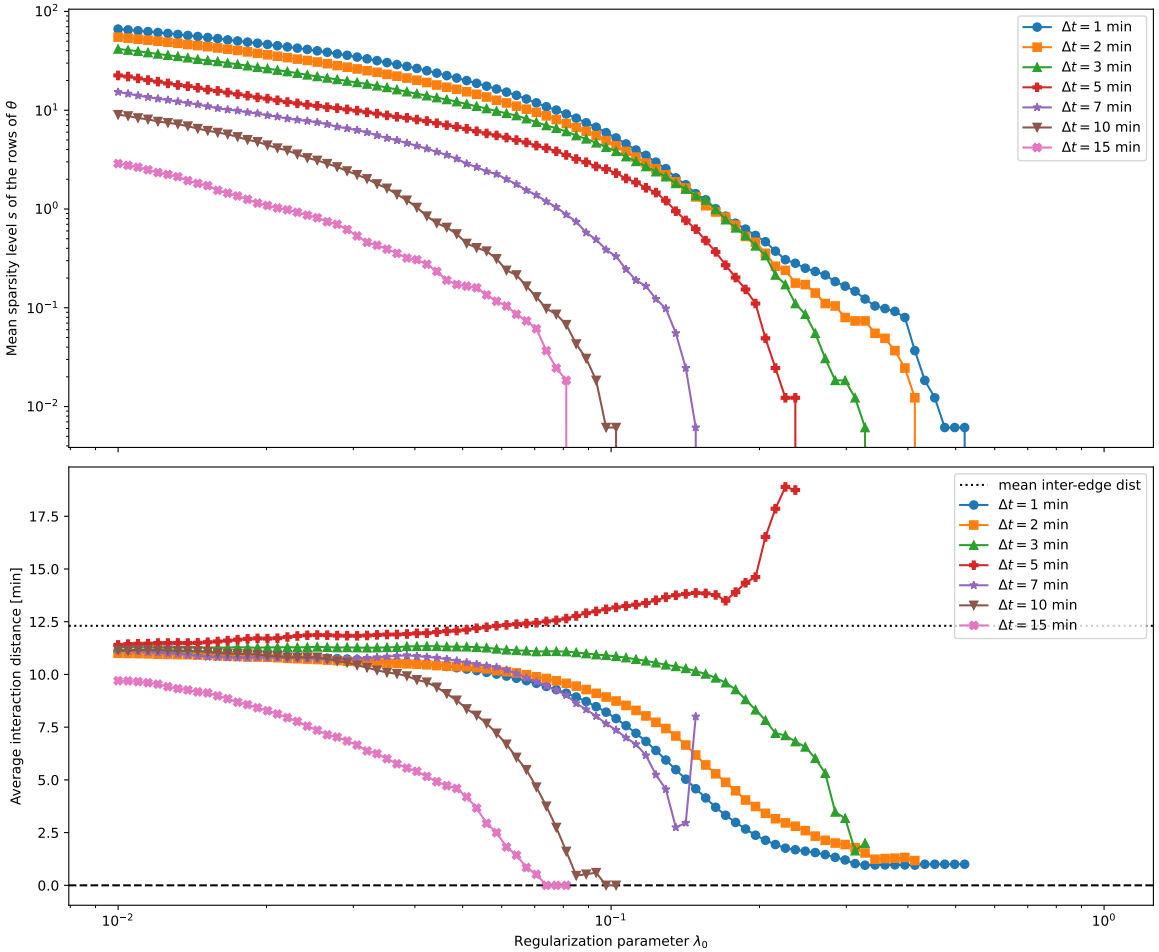

Figure 11: Effect of regularization and time discretization interval on some features of the estimate $\widehat{\theta}$

Improving the model

- Capture the congestion with a graph Gaussian Process [9] to
  - Add nonlinearities
  - Switch to continuous time
- Consider external features when predicting arrival times
- Propose a learning procedure based on the EM algorithm [5]

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The Flatland challenge[1]

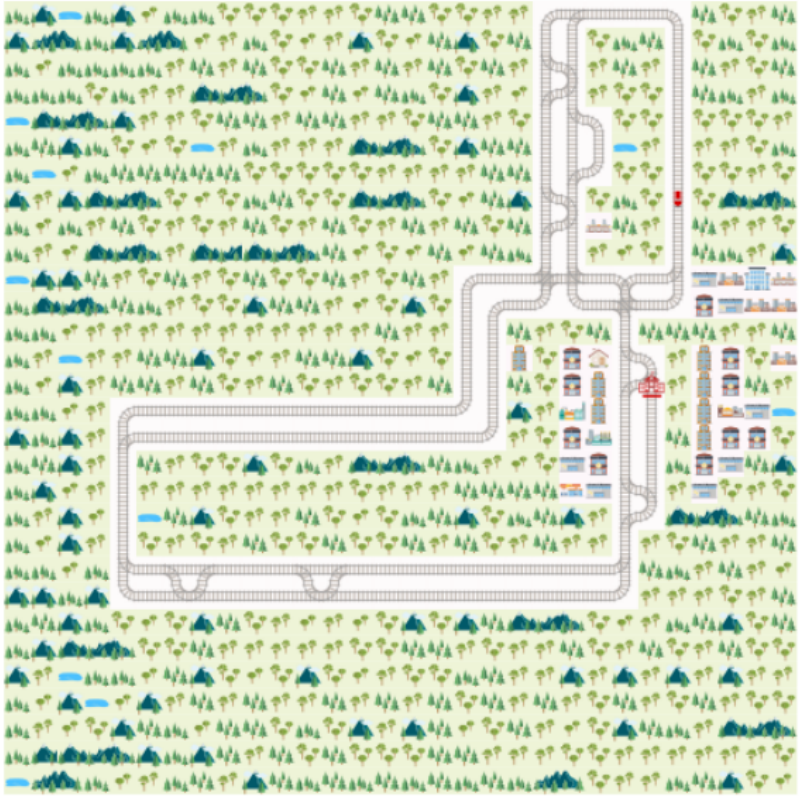

Figure 12: Example Flatland map [6]

- Flatland provides a simulation and evaluation tool for real-time railway traffic management
- Goal: reschedule trains to minimize delays in spite of stochastic incidents

[1] https://www.aicrowd.com/challenges/flatland-3

A simple policy

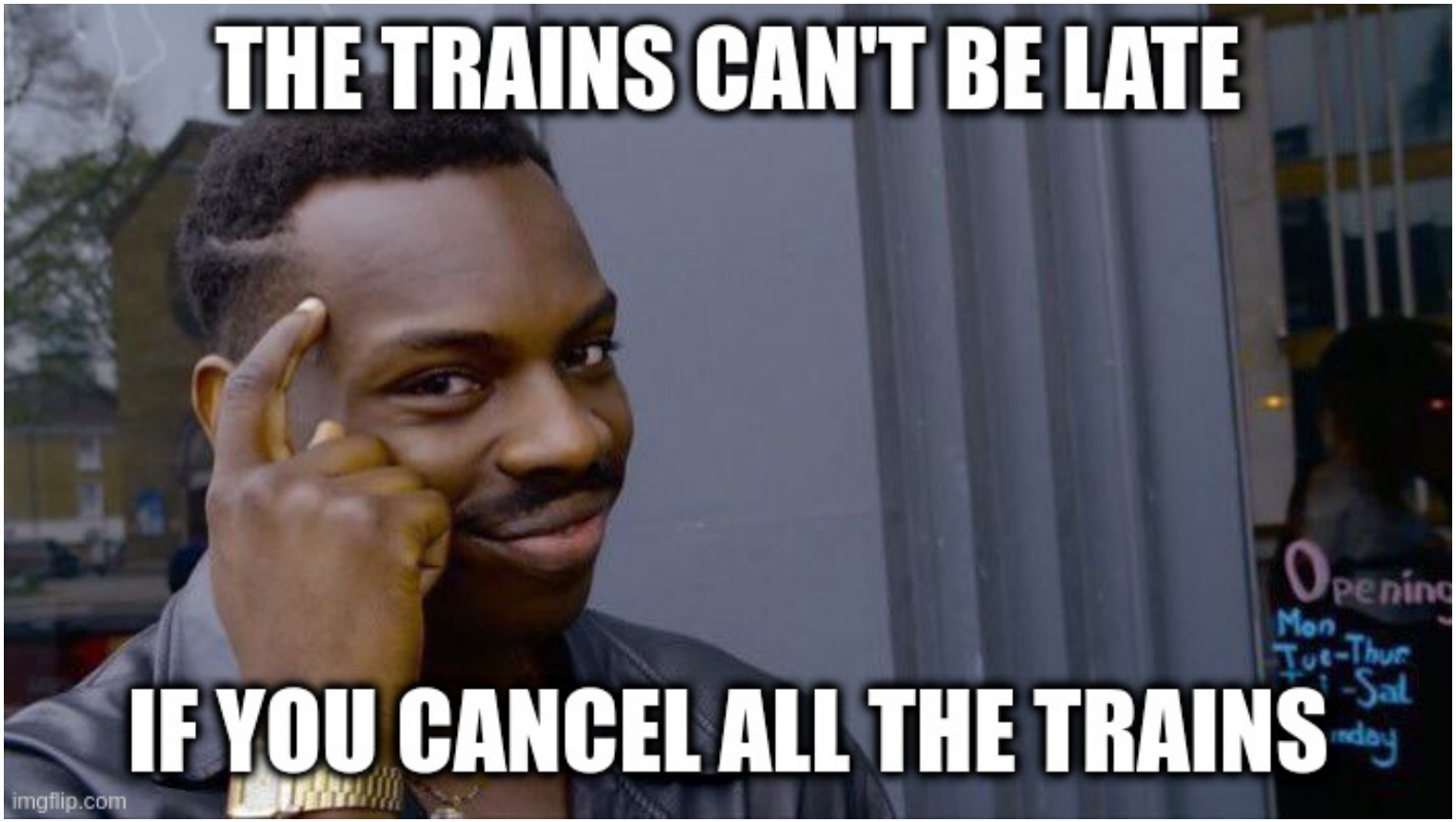

Figure 13: On the importance of carefully selecting the objective function

Simulation environment

- Railway networks are generated from 8 cell templates
- Trains are given an initial and target cell & a time window in which they must complete their journey

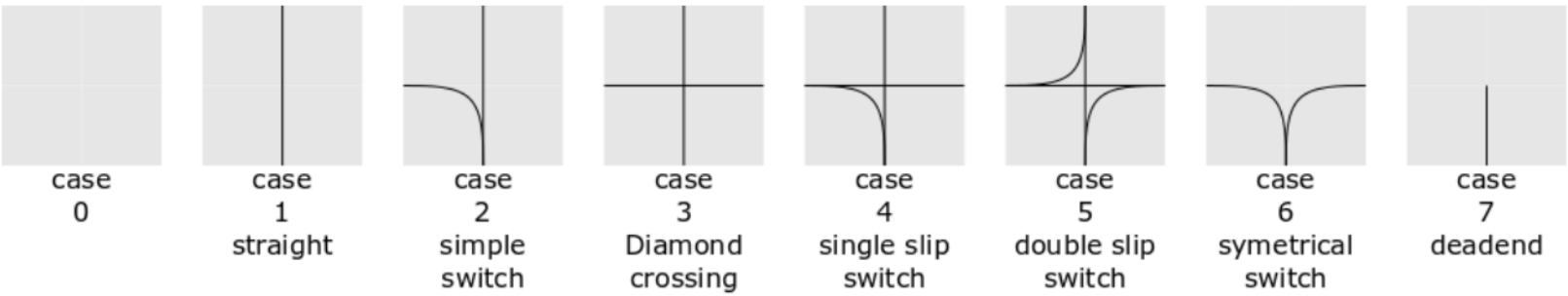

Figure 14: Flatland cell templates [6]

If it were easy, someone else would do it

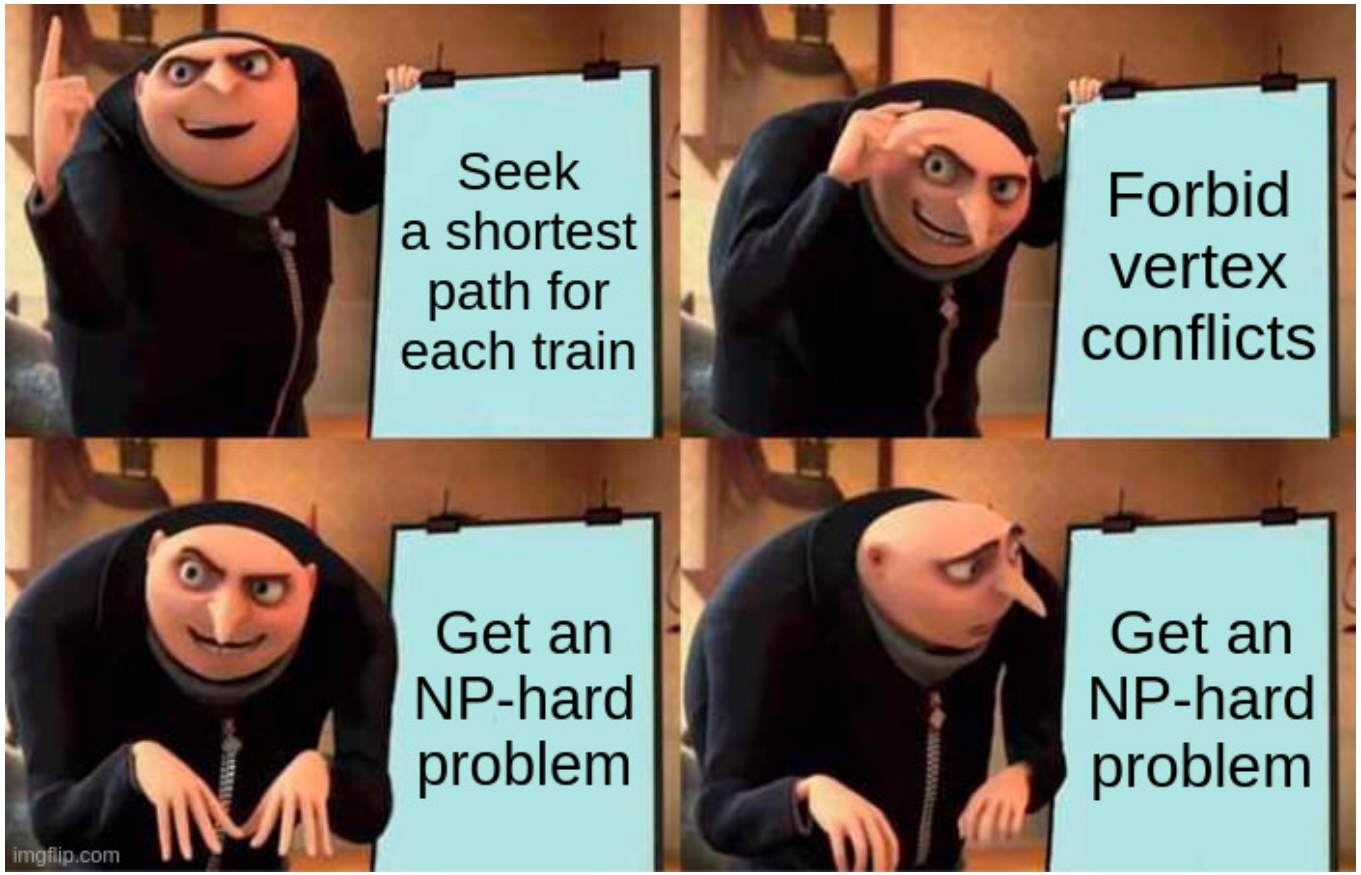

Figure 15: Theorem: Any interesting real-life problem is intractable

Why is it so difficult?

- The Multi-Agent Path Finding (MAPF) problem [12] is already NP-hard (basically a multiflow)
- Flatland introduces additional hurdles:
  - Multiple speed profiles
  - Random malfunctions following a Poisson process
- We must continuously adapt the initial planning to prevent slowdowns and deadlocks: need for a fast algorithm

Prioritized Planning (PP)

- The simplest way to solve deterministic MAPF [11]:
  1. Rank the trains in an arbitrary order
  2. For each one, find a shortest path that doesn't conflict with higher-ranking trains
- Drawbacks: greedy, doesn't always find a solution, let alone an optimal one
- What if we could improve it with machine learning using [8]? To be continued...

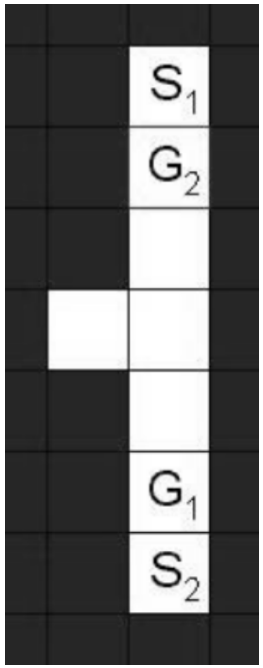

Figure 16: PP is not complete [10]

That's all, folks!

Any questions?

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